

CreditIQ - AI Powered Credit Risk Underwriting Platform

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Table of Contents

1. Problem Statement
2. Dataset Overview
3. System Architecture
4. Machine Learning Pipeline
5. Executive Dashboard
6. Risk Prediction
7. Explainability

8. Decision Rules
9. AI Analyst
10. Results
11. Key Features
12. Future Scope

Problem Statement

Challenges in Traditional Underwriting

- Manual and time-consuming decisions
- Limited transparency in risk assessment
- Difficulty scaling across large applicant volumes
- Inconsistent decision-making,

Solution

AI-powered platform for:

- ✓ Risk Prediction
- ✓ Explainability
- ✓ Decision Rules
- ✓ Portfolio Analytics

Dataset Overview

Home Credit Default Risk Dataset

- 307,511 Applications

Target:

- Loan Default Prediction

Default Rate:

- 8.1%

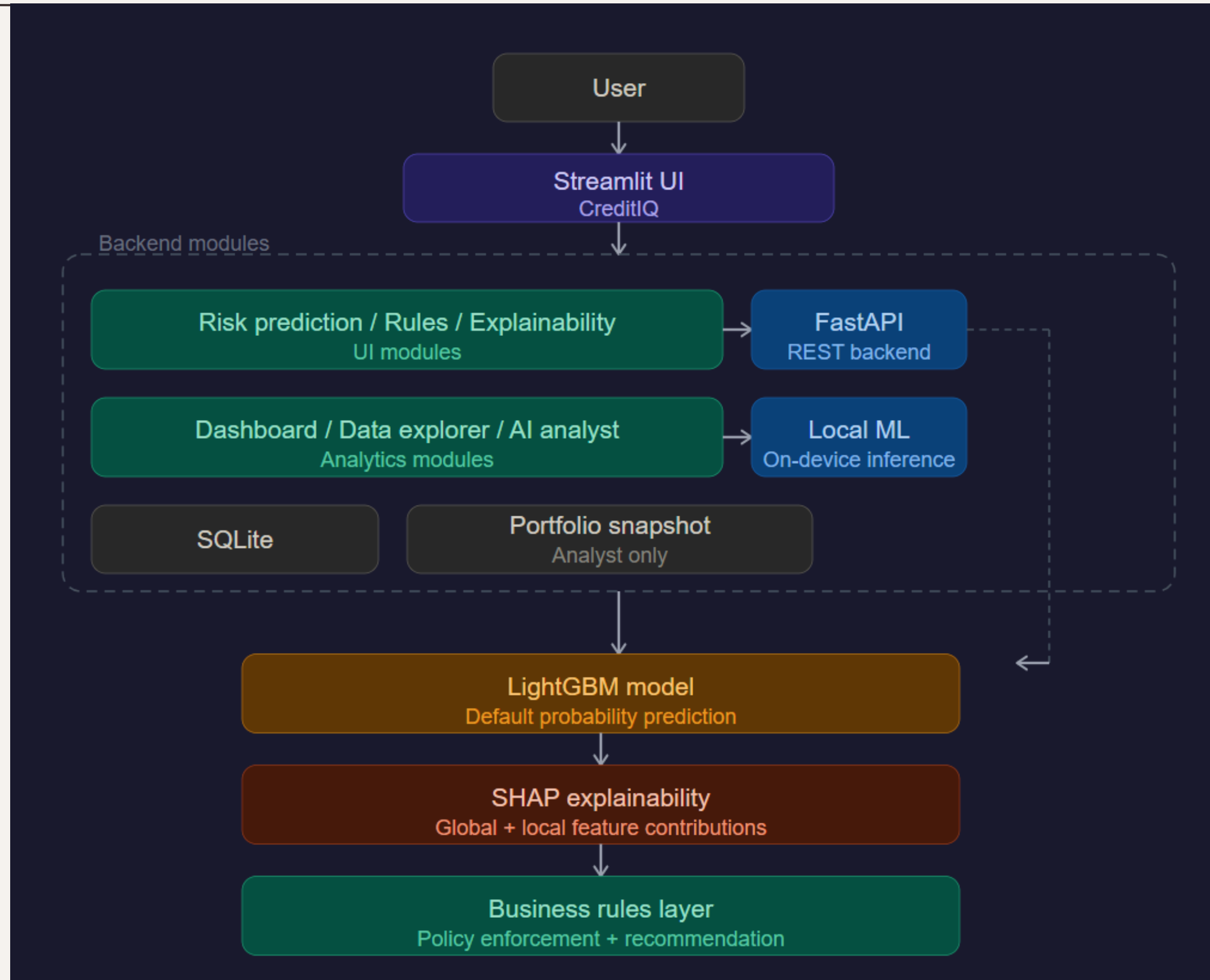
Model:

- LightGBM

Key Features:

- Credit Scores
- Income Information
- Loan Details
- Employment History

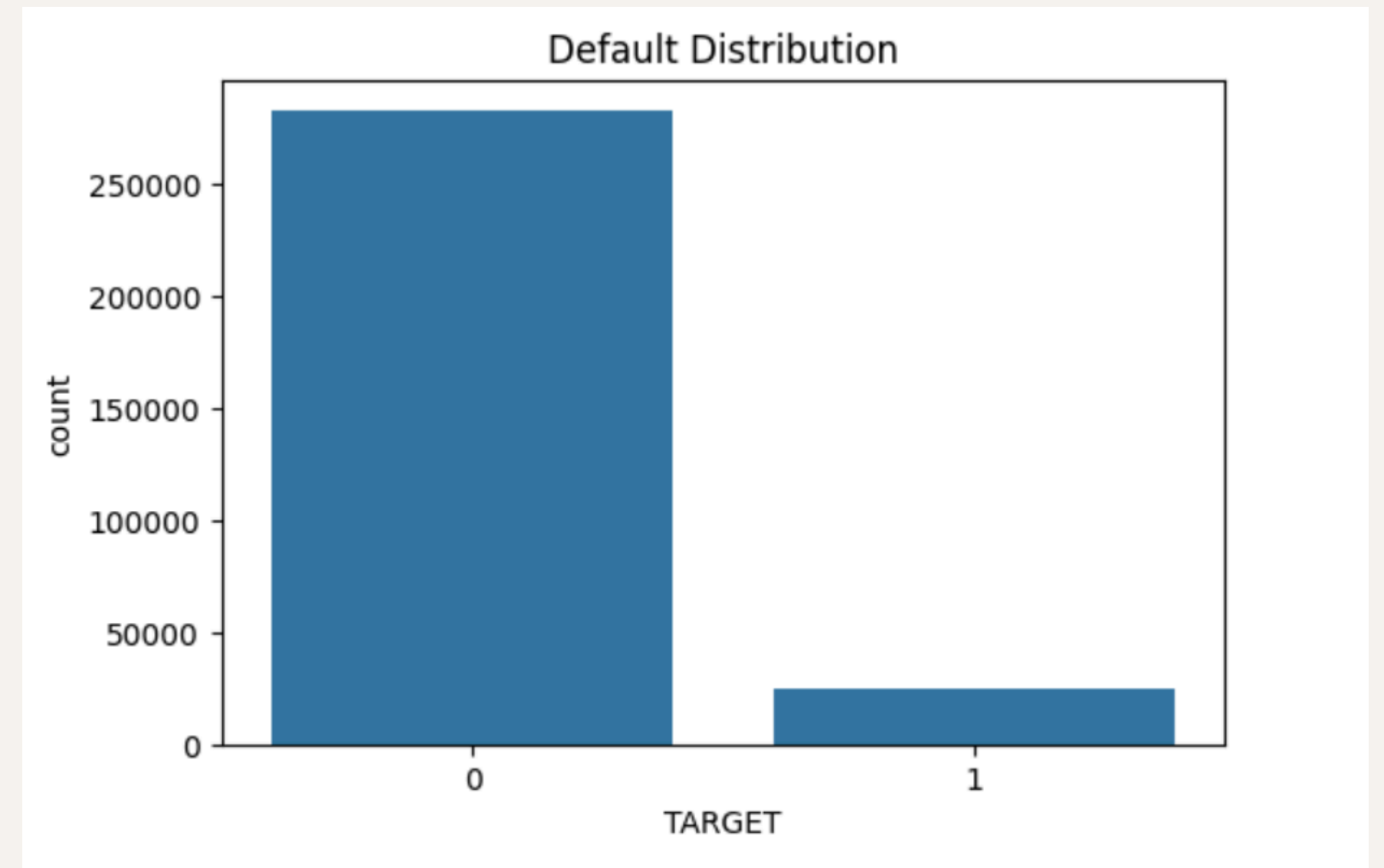
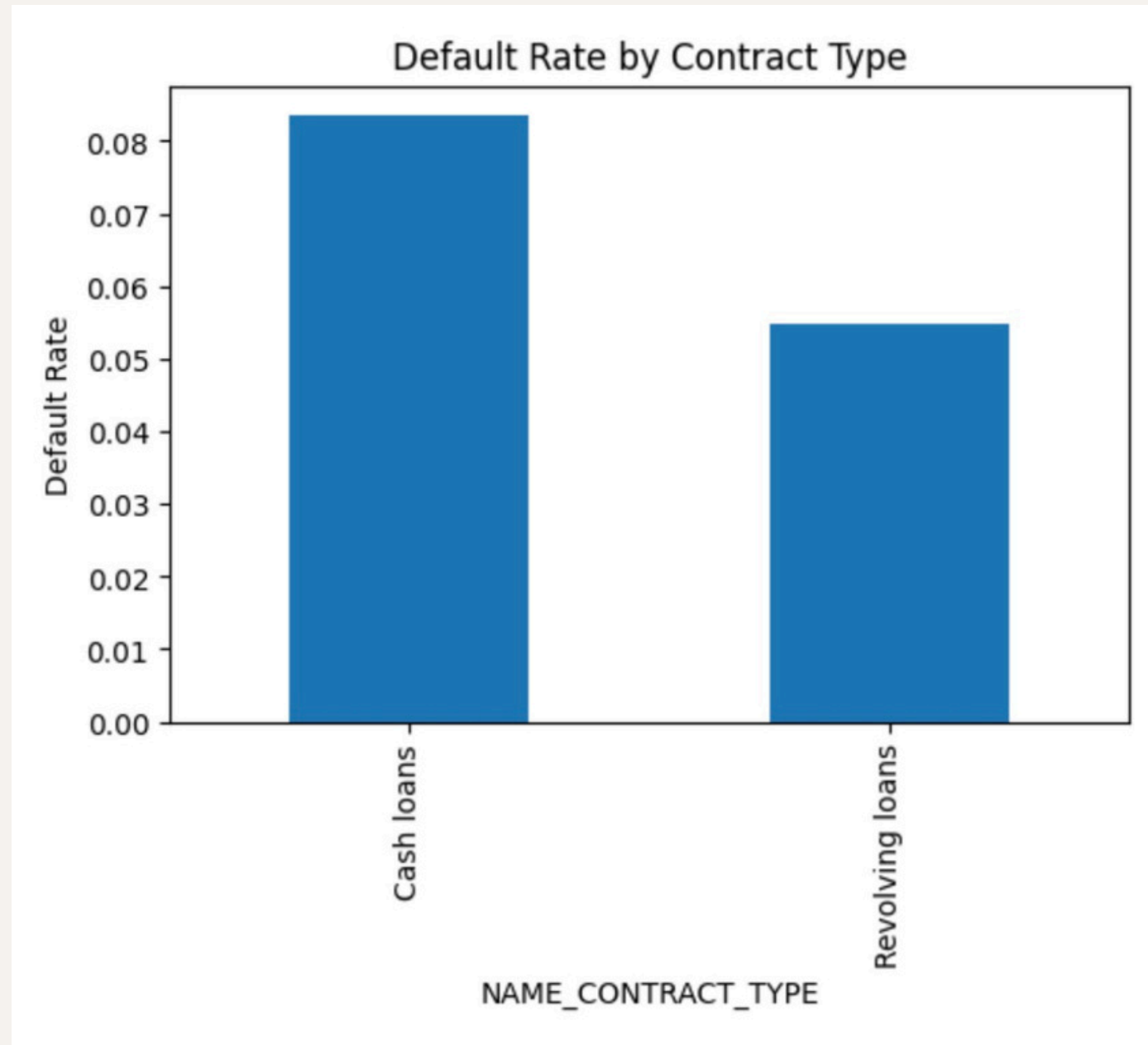
System Architecture



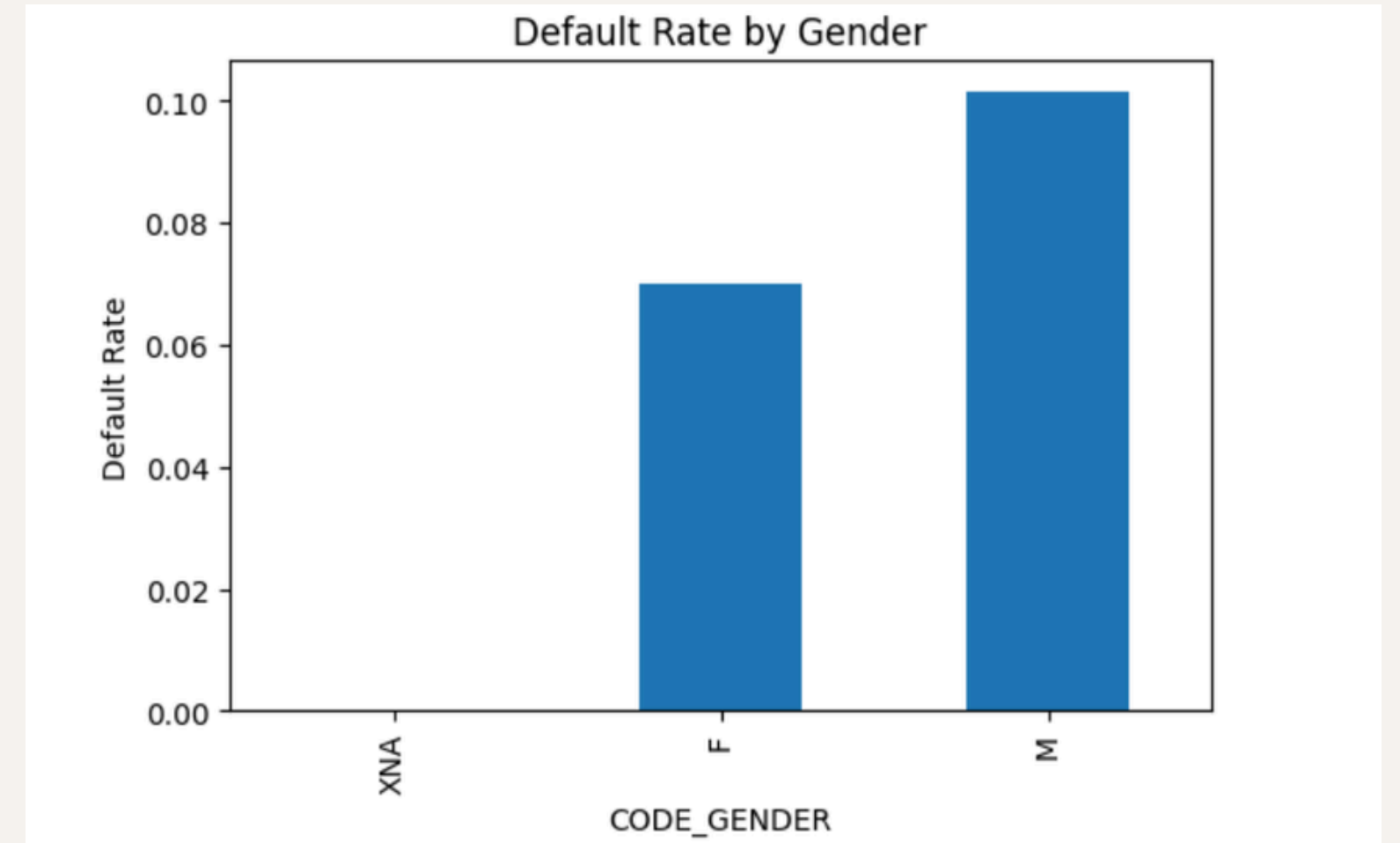
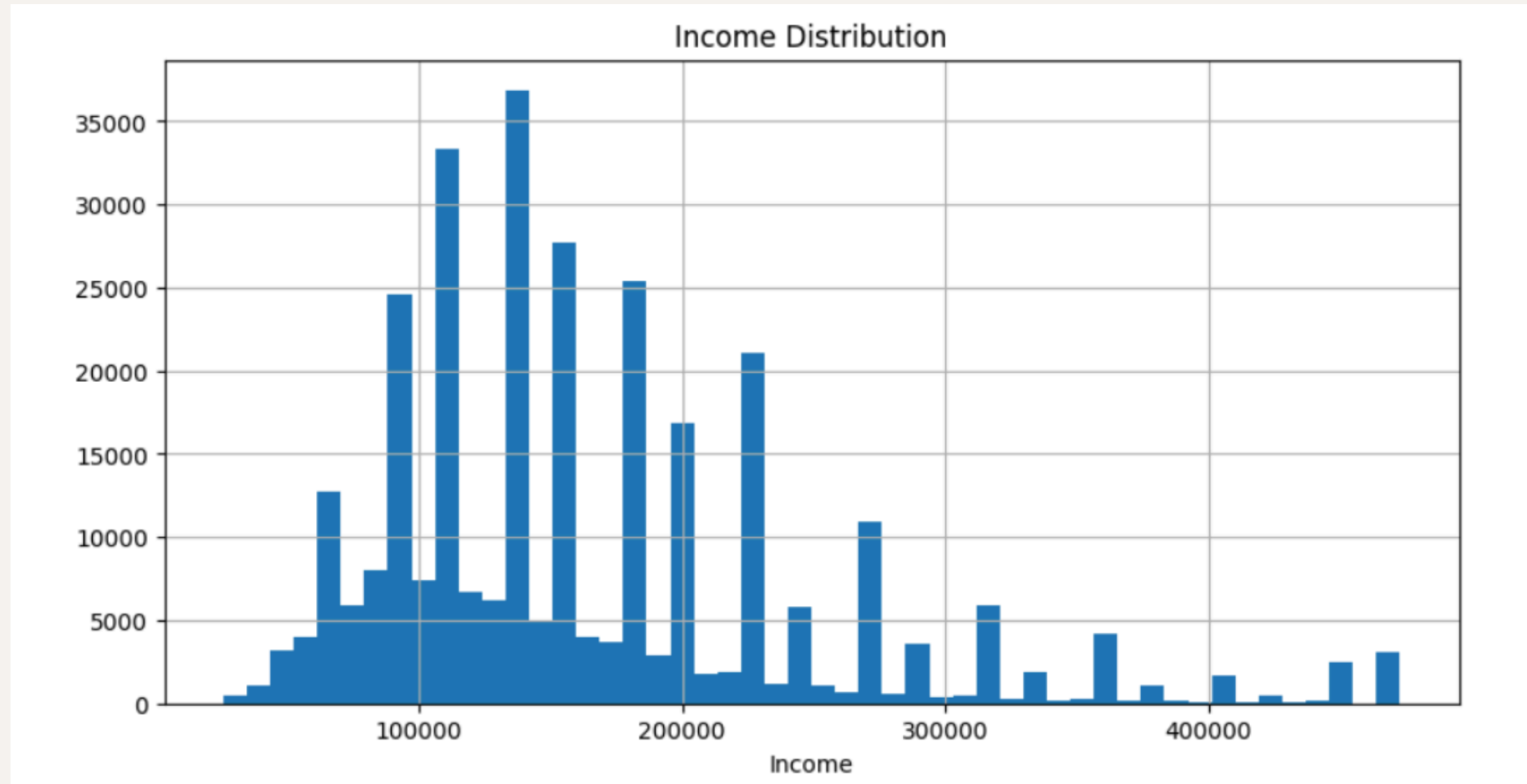
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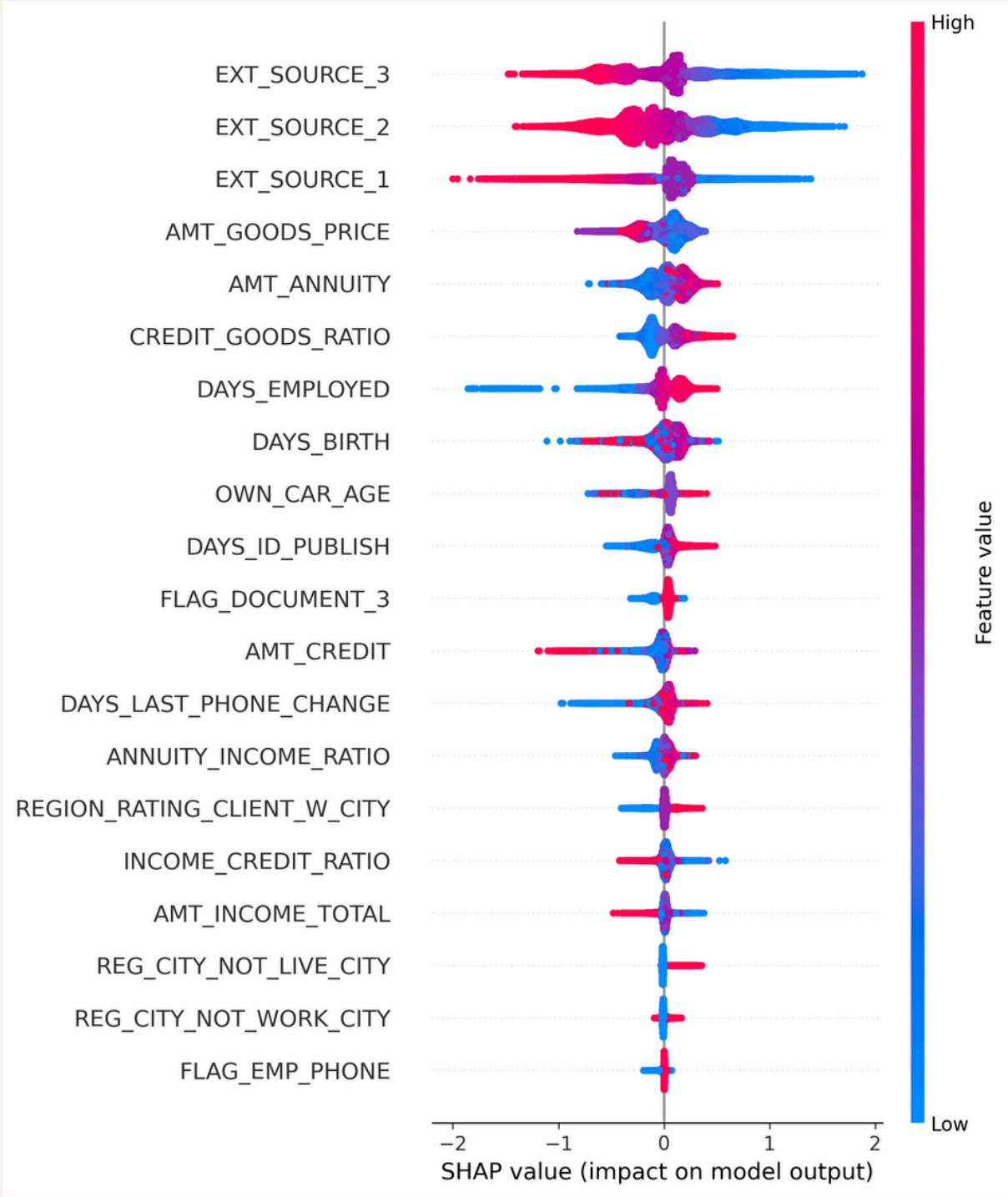
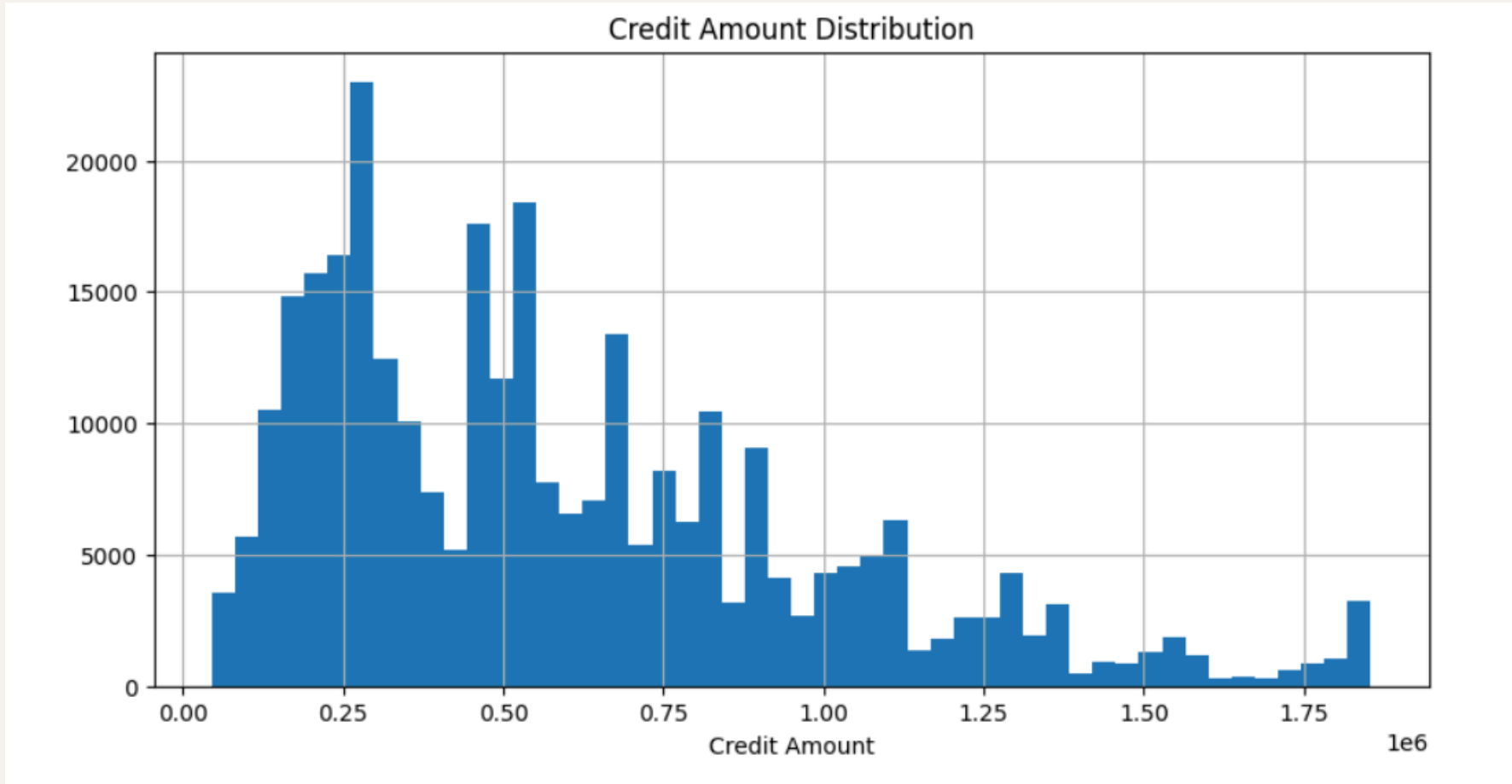
EDA Analysis



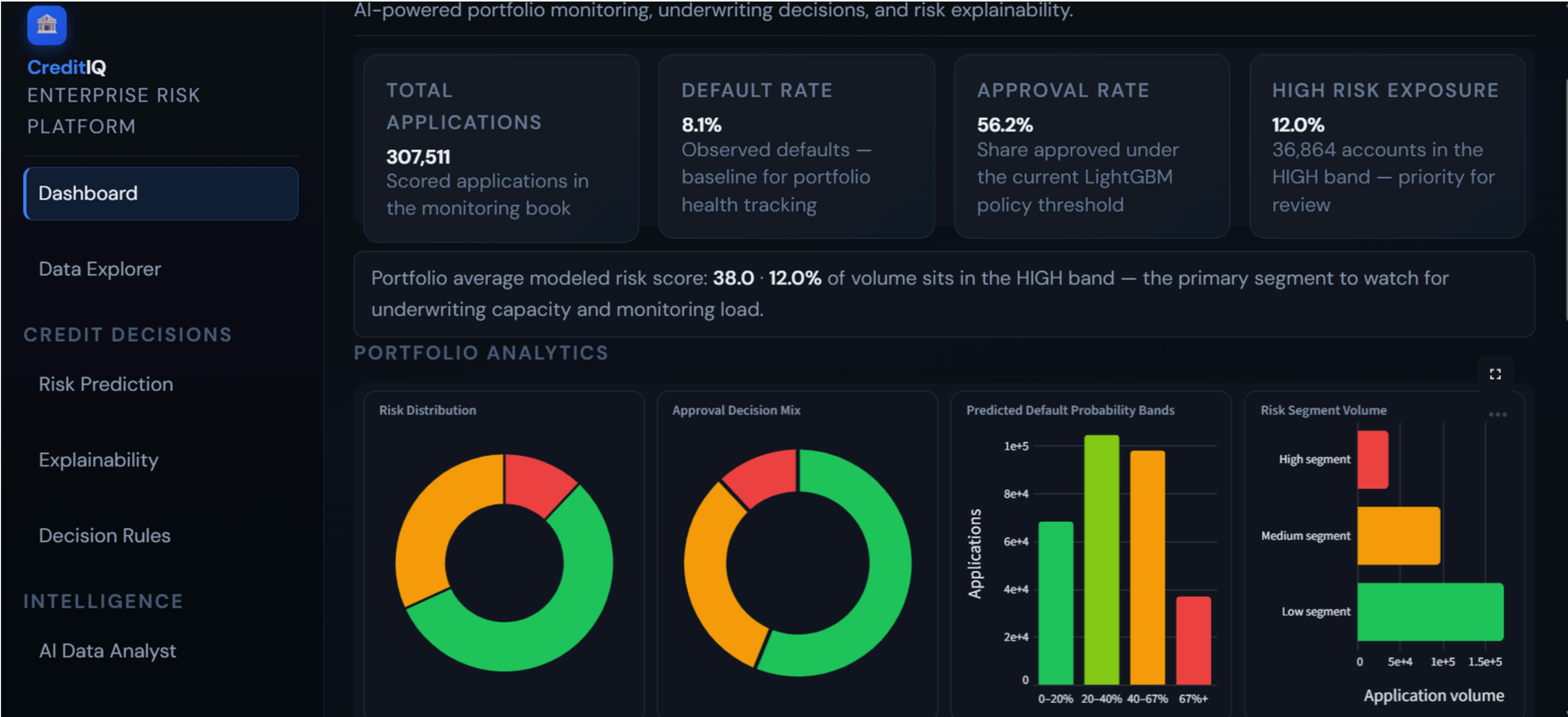
EDA Analysis



EDA Analysis



Executive Dashboard



Executive Dashboard



Risk Prediction

**CreditIQ**
ENTERPRISE RISK
PLATFORM

Dashboard

Data Explorer

CREDIT DECISIONS

Risk Prediction

Explainability

Decision Rules

INTELLIGENCE

AI Data Analyst

APPLICANT IDENTITY

Applicant name

Priya

Applicant ID

APP-001

FINANCIAL PROFILE

Annual income

864,000

Monthly annuity (₹/mo)

18,500

Credit amount

450,000

Goods price

450,000

BUREAU SCORES

External credit score 1

0.52

External credit score 2

0.61

External credit score 3

0.68

EMPLOYMENT & REGION

Employment years

4.0

Region rating

2 — Standard

✓ Scored in 3167 ms

Low Risk

19

RISK SCORE

DEFAULT PROBABILITY

18.8%

APPROVAL PROBABILITY

81.2%

CONFIDENCE

99%

RECOMMENDATION

✓ Approve

View Explanation →

12

Risk Prediction



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ENTERPRISE RISK PLATFORM

Dashboard

Data Explorer

CREDIT DECISIONS

Risk Prediction

Explainability

Decision Rules

INTELLIGENCE

AI Data Analyst

APPLICANT IDENTITY

Applicant name

Rahul

Applicant ID

APP-002

FINANCIAL PROFILE

Annual income

864,00

Monthly annuity (₹/mo)

18,500

Credit amount

450,000

Goods price

450,000

BUREAU SCORES

External credit score 1

0.52

External credit score 2

0.31

External credit score 3

0.28

EMPLOYMENT & REGION

Employment years

2

Region rating

2 — Standard

✓ Scored in 146 ms

Medium Risk

57

RISK SCORE

DEFAULT PROBABILITY

56.7%

APPROVAL PROBABILITY

43.3%

CONFIDENCE

86%

RECOMMENDATION

Manual review

View Explanation →

13

Explainability

**CreditIQ**
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PLATFORM

Dashboard

Data Explorer

CREDIT DECISIONS

Risk Prediction

Explainability

Decision Rules

INTELLIGENCE

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Explainability

Local SHAP drivers and underwriting context for the selected applicant.

☒ Use latest Risk Prediction assessment

☐ Explain custom applicant

Applicant: Priya · ID APP-001

RISK SCORE
19
LOW RISK

RECOMMENDATION
Approve
APPROVE

DEFAULT PROBABILITY
18.8%
Model-estimated default likelihood

AI SUMMARY

This applicant is assessed as low risk, primarily influenced by employment history and id document. Strong external credit history partially offsets default risk.

RISK DRIVERS

- Employment history
- ID document
- ID tenure

POSITIVE FACTORS

- Strong external credit history
- Vehicle age
- External credit score 1

Explainability

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Dashboard

Data Explorer

CREDIT DECISIONS

Risk Prediction

Explainability

Decision Rules

INTELLIGENCE

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ENTERPRISE RISK PLATFORM

Explainability

Local SHAP drivers and underwriting context for the selected applicant.

☐ Use latest Risk Prediction assessment

☒ Explain custom applicant

External credit score 2

0.61

-

+

External credit score 1

0.52

-

+

External credit score 3

0.68

-

+

Credit amount (₹)

0.00

-

+

Annual income (₹)

0.00

-

+

Employment years

4.00

-

+

Generate explanation

Decision Rules

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Dashboard

Data Explorer

CREDIT DECISIONS

Risk Prediction

Explainability

Decision Rules

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ACTIVE RULES
7
0 TRIGGERED

HIGH RISK
3
DECLINE THRESHOLD

MEDIUM RISK
2
REVIEW THRESHOLD

LOW RISK
2
APPROVE THRESHOLD

How these rules work

FILTER BY BAND

☒ All

☐ High Risk (3)

☐ Medium Risk (2)

☐ Low Risk (2)

HIGH RISK

Weak bureau score + high credit exposure

CONDITIONS

External Credit Score 2 < 0.35

Requested Loan Amount > ₹500,000

REASON: Low bureau score and high borrowing exposure

CONFIDENCE 91% COVERAGE 4.0% PRECISION 88% LIFT 2.4x

ACTION

X Decline

HIGH RISK

Short tenure + low income

Decision Rules



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Dashboard

Data Explorer

CREDIT DECISIONS

Risk Prediction

Explainability

Decision Rules

INTELLIGENCE

AI Data Analyst

EMI-to-Income Ratio > 0.25

Income-to-Loan Ratio < 0.20

REASON: Elevated payment burden and thin income coverage

CONFIDENCE 80%

COVERAGE 1.8%

PRECISION 76%

LIFT 1.5x

Manual Review

LOW RISK

Strong bureau scores — fast-track approve

CONDITIONS

External Credit Score 2 ≥ 0.70

External Credit Score 3 ≥ 0.65

REASON: Strong bureau performance

CONFIDENCE 94%

COVERAGE 7.1%

PRECISION 96%

LIFT 0.3x

ACTION

✓ Approve

LOW RISK

Stable employment + affordable annuity

CONDITIONS

Employment Tenure ≥ 5 years

EMI-to-Income Ratio < 0.25

REASON: Stable employment and affordable payment load

CONFIDENCE 92%

COVERAGE 6.0%

PRECISION 94%

LIFT 0.4x

ACTION

✓ Approve

Decision Rules



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PLATFORM

Dashboard

Data Explorer

CREDIT DECISIONS

Risk Prediction

Explainability

Decision Rules

INTELLIGENCE

AI Data Analyst

HIGH RISK

Low external score bundle

CONDITIONS

› External Credit Score 1 < 0.30

› External Credit Score 3 < 0.40

REASON: Multiple weak external credit indicators

CONFIDENCE 86%

COVERAGE 2.3%

PRECISION 83%

LIFT 2.0x

ACTION

X Decline

MEDIUM RISK

Annuity stress + weaker region rating

CONDITIONS

› EMI-to-Income Ratio > 0.30

› Region Risk Rating = 3

REASON: Elevated payment burden and weaker region risk

CONFIDENCE 82%

COVERAGE 2.0%

PRECISION 79%

LIFT 1.7x

ACTION

Manual Review

MEDIUM RISK

Elevated payment burden

CONDITIONS

› EMI-to-Income Ratio > 0.25

› Income-to-Loan Ratio < 0.20

ACTION

Manual Review

AI Data Analyst



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PLATFORM

Dashboard

Data Explorer

CREDIT DECISIONS

Risk Prediction

Explainability

Decision Rules

INTELLIGENCE

AI Data Analyst

CONVERSATION

SUGGESTED QUESTIONS

Approval Rate

Default Rate

Portfolio Summary

Average Income

Loan Type Risk

Risk By Gender

Top Risk Regions

You

Which loan type has highest default rate?

Analyst

Cash loans shows the highest default rate at **8.3%**. Cash loans typically warrant tighter underwriting and monitoring relative to other products.

QUERY WORKSPACE

Generated SQL

```
SELECT NAME_CONTRACT_TYPE,
       ROUND(100.0 * AVG(TARGET), 2) AS default_rate_pct,
       COUNT(*) AS applications
FROM application_train
GROUP BY NAME_CONTRACT_TYPE
ORDER BY default_rate_pct DESC;
```

Results

NAME_CONTRACT_TYPE	default_rate_pct	applications
Cash loans	8.35	278232
Revolving loans	5.48	29279

BUSINESS INSIGHT

Cash loans shows the highest default rate at **8.3%**. Cash loans typically warrant tighter underwriting and monitoring relative to other products.

AI Data Analyst

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Approval Rate

Default Rate

Portfolio Summary

Average Income

Loan Type Risk

Risk By Gender

Top Risk Regions

The portfolio approval rate is **56.2%** under the current LightGBM underwriting policy (batch-scored recommendations). Applicants classified as APPROVE are cleared for origination; MEDIUM-risk cases are routed to manual review.

You
Show risk by gender

Analyst

Default rates by gender — M: 10.1% default (105,059 applications); F: 7.0% default (202,448 applications); XNA: 0.0% default (4 applications).

```
SELECT CODE_GENDER,
COUNT(*) AS applications,
ROUND(100.0 * AVG(TARGET), 2) AS default_rate_pct
FROM application_train
GROUP BY CODE_GENDER
ORDER BY default_rate_pct DESC;
```

Results

CODE_GENDER	applications	default_rate_pct
M	105059	10.14
F	202448	7
XNA	4	0

BUSINESS INSIGHT

Default rates by gender — M: 10.1% default (105,059 applications); F: 7.0% default (202,448 applications); XNA: 0.0% default (4 applications).

Results

Model Results

- ROC-AUC: 0.7516
- Precision: 25.66%
- Recall: 37.42%
- F1 Score: 30.45%

Portfolio Metrics

- 307,511 Applications
- 56.2% Approval Rate
- 12.0% High-Risk Exposure

Future Scope

- Model Monitoring
- Drift Detection
- Advanced Conversational Analytics
- Cloud Deployment
- Automated Retraining

CreditIQ demonstrates an end-to-end AI underwriting workflow combining Machine Learning, Explainable AI, and Business Intelligence.

THANK YOU